

LAB 04

Stilt walking

Stilt Walking: Adapting Control to
Changed Contact Geometry

Environment design · PPO · Honest verification

BUILD → TRAIN → VERIFY → EXPLAIN



What you will build — and what the evidence says

- 01 Use case: Teach morphology-aware simulation, staged PPO continuation, command-directed walking gates, and why a compatible tensor interface is not a hardware safety claim.
- 02 By the end: describe the scene, map observations to actions, explain the reward, trace one PPO update, run a bounded experiment, and defend an evaluation claim.
- 03 The evidence covers one fixed 2 cm morphology, blend 0, 0.014 kg mass, and one 0.25 m/s command; morphology generalization is unknown.
- 04 Scope: local MuJoCo simulation. Do not deploy these classroom artifacts to a physical robot; hardware transfer requires a separate validated process.

Materials and prerequisites · 1/2

- 01 Apple Silicon Mac with installed RLX/MLX, MuJoCo, ONNX Runtime, rendering dependencies, and Node.
- 02 Tracked base and refinement recipes: `stilts-base.json` and `stilts.json`.
- 03 Procedurally generated rigid stilt bodies, mesh collisions, moved foot sites, and raised STAND keyframe.
- 04 Saved base checkpoint plus adjacent normalization sidecar; both are required for refinement.

Materials and prerequisites · 2/2

05

Final checkpoint, ONNX actor, evaluation JSON, five-seed traces, optimizer JSONL, MP4, contact sheet, and render receipt.

Environment and policy contract · 1/2

- 01 Nominal XML MuJoCo at 50 Hz with 2 cm stilts, blend 0, and 0.014 kg added mass per foot.
- 02 Original sole collisions are disabled; each ankle receives a rigid stilt mesh, new collision, bottom foot site, and raised standing root.
- 03 Observation remains float32 batch x 61; stilt height, blend, mass, contacts, world position, and absolute yaw are not actor inputs.
- 04 Action remains float32 batch x 14 raw target offsets; raw values are observed and penalized, while applied offsets clip to [-4,4].

Environment and policy contract · 2/2

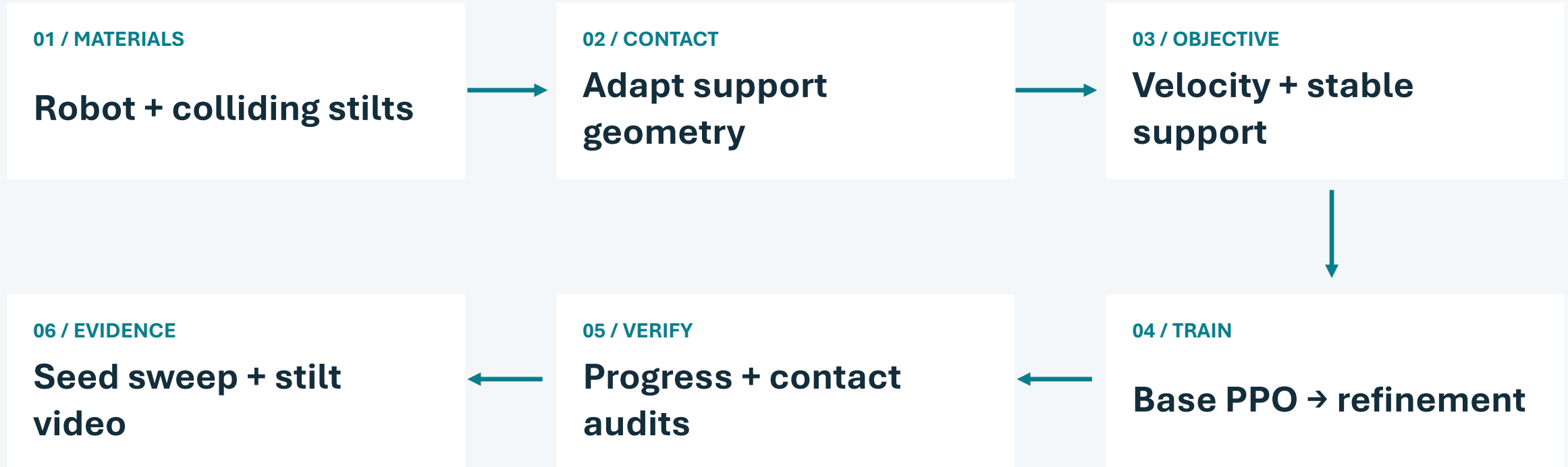
05

Fixed twist command is $[0.25, 0, 0]$; episode horizon is 500 steps or 10 seconds.

06

Domain randomization, observation noise, action delay, and random yaw are disabled; the policy specializes to one morphology.

How this case is built: one evidence loop



Failure is data: change one cause, retrain, and repeat the same evaluation gates.

Build the case step by step · 1/2

- 01 Build the morphology before environment creation: disable sole collisions, attach two rigid stilt bodies, move contact sites, and raise STAND by 0.02 m.
- 02 Train Stilt v2 from random actor and critic weights for 6,000,640 transitions at a fixed 0.25 m/s command.
- 03 Retain the v2 failure: it learned alternating locomotion, but four of five historical episodes failed signed intended displacement.
- 04 Copy checkpoint and .safetensors.json sidecar into a fresh v3 run; preserve actor, critic, log std, and normalization lineage.

Build the case step by step · 2/2

05

Refine for 1,048,576 transitions while increasing angular tracking from 0.5 to 4 and leaving morphology and gate unchanged.

06

Export deterministic ONNX and evaluate complete 10-second episodes with intended displacement and bilateral contact-event criteria.

07

Audit seeds 501-505 against zero, random, and initial controls; validate ONNX parity and inspect the complete stilt MP4.

PPO: improve the actor without trusting a huge step

$$\text{ratio} = \pi_{\text{new}}(a \mid o) / \pi_{\text{old}}(a \mid o)$$

Maximize $\text{mean}[\min(\text{ratio} \times A, \text{clip}(\text{ratio}, 1 - \epsilon, 1 + \epsilon) \times A)]$

Actor: chooses joint actions. Critic: estimates future discounted reward.

A = advantage. ϵ = clipping range. Old log probabilities stay fixed for each collected batch.

Follow one rollout through the learner

- 01 Collect observations, sampled actions, old log probabilities, values, rewards and episode-boundary flags using the current stochastic actor.
- 02 Compute TD residuals and generalized advantage estimates (GAE). A true terminal has zero continuation value; a timeout may bootstrap the saved final observation.
- 03 Cut advantage traces at reset boundaries. Never bootstrap a timeout from the first observation of the next episode.
- 04 Reuse this batch for PPO minibatch updates, then collect fresh on-policy data. Export the deterministic actor and its observation transform for evaluation.

Paper exercise: count samples and clip the objective

- 01 Illustrative arithmetic, NOT this run: $16 \text{ environments} \times 128 \text{ steps} = 2,048$ fresh transitions per rollout. Four epochs reuse those transitions; they do not create 8,192 new samples.
- 02 With $\epsilon = 0.2$, ratio = 1.4 and $A = +2$: $\min(2.8, 2.4) = 2.4$. The minimized policy term is -2.4 before averaging and additional loss terms.
- 03 With the same ratio and $A = -2$: $\min(-2.8, -2.4) = -2.8$. Explain why clipping is asymmetric once the sign of the advantage changes.
- 04 Run from repo root: `python3 docs/microduck-training-student-book/learning_lab.py --gae`. This is a deterministic arithmetic lab, not new policy training.

Reward design: what the agent is paid to do · 1/2

- 01 track_lin_vel 20: Gaussian body-frame command tracking including vertical-speed error.
- 02 track_ang_vel 4: Gaussian yaw-command tracking plus roll/pitch angular-rate costs; base weight was 0.5.
- 03 upright 1: bounded projected-gravity posture score; pose remains present with weight 0.
- 04 feet_air_time 4: dense per-foot reward for air timers inside the walking window.

Reward design: what the agent is paid to do · 2/2

05

head_pose 0.1 tracks head command; ang_vel_xy_penalty 0.02 is non-positive by construction.

06

Inherited action-rate penalty is rescaled by 0.2 and follows an environment-lifetime schedule up to effective coefficient 0.20.

Recorded training recipe · 1/2

- 01 Base v2: seed 7, 6,000,640 transitions, 16 envs x 128 steps, 2,930 rollouts, 46,880 optimizer minibatches, learning rate 0.0003.
- 02 Continuation v3: 1,048,576 transitions, 512 rollouts, 8,192 optimizer minibatches, learning rate 0.0001.
- 03 Entire selected lineage: 7,049,216 transitions, 3,442 rollout/update cycles, and 55,072 optimizer minibatches.
- 04 Both stages use 4 minibatches x 4 epochs, gamma 0.99, GAE lambda 0.95, clip 0.2, value coefficient 1, entropy 0, and max grad norm 0.5.

Recorded training recipe · 2/2

05

Base initial std is 0.6; continuation loads learned actor, critic, log std, observation stats, and return stats but starts fresh Adam state.

06

Update log fields are phase, steps, seconds, optimizer_steps, mean_loss, policy_loss, value_loss, entropy, approximate_kl, clip_fraction, explained_variance, env_steps.

Hands-on commands: inspect, smoke, train · 1/3

01

```
From .:  
./scripts/train-stilts.sh --dry-run  
Inspect the guarded two-stage Stilt pipeline without training.
```

02

```
From .:  
./scripts/train-stilts.sh  
Run the independent Full Stilt base and refinement pipeline with fresh outputs.
```

Hands-on commands: inspect, smoke, train · 2/3

03

From .:

```
RUN=stilts-classroom-$(date +%Y%m%d-%H%M%S); ./scripts/train-stilts.sh --recipe-json docs/remaining-scenarios-e2e/recipes/stilts.json --init-from rlx/runs/studio/stilts/stilts-e2e-20260907-v2/stilts.safetensors --run-name "$RUN"
```

Refine the retained v2 checkpoint with its adjacent sidecar in a fresh run; keep RUN in this terminal for the following audit. Requires the ignored historical base artifacts.

04

From .:

```
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/recipes/stilts.json --run "rlx/runs/studio/stilts/${RUN:?Run the refinement command first in this shell}" --output "rlx/artifacts/$RUN-audit" --seeds 501 502 503 504 505 --render
```

Audit a completed Stilt reproduction over five held-out reset seeds.

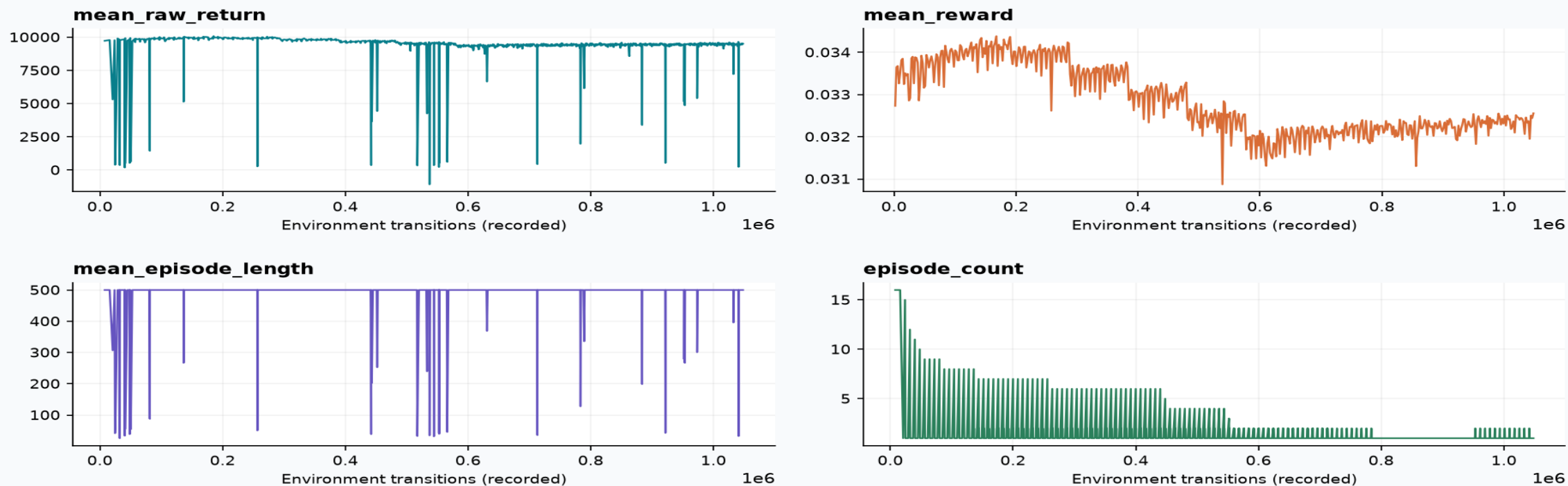
Hands-on commands: inspect, smoke, train • 3/3

05

```
From .:  
python3 -m unittest discover -s scripts/scenarios -p 'test_*.py'  
Run scenario-pipeline unit tests without starting training.
```

Reward: does the logged signal match the task?

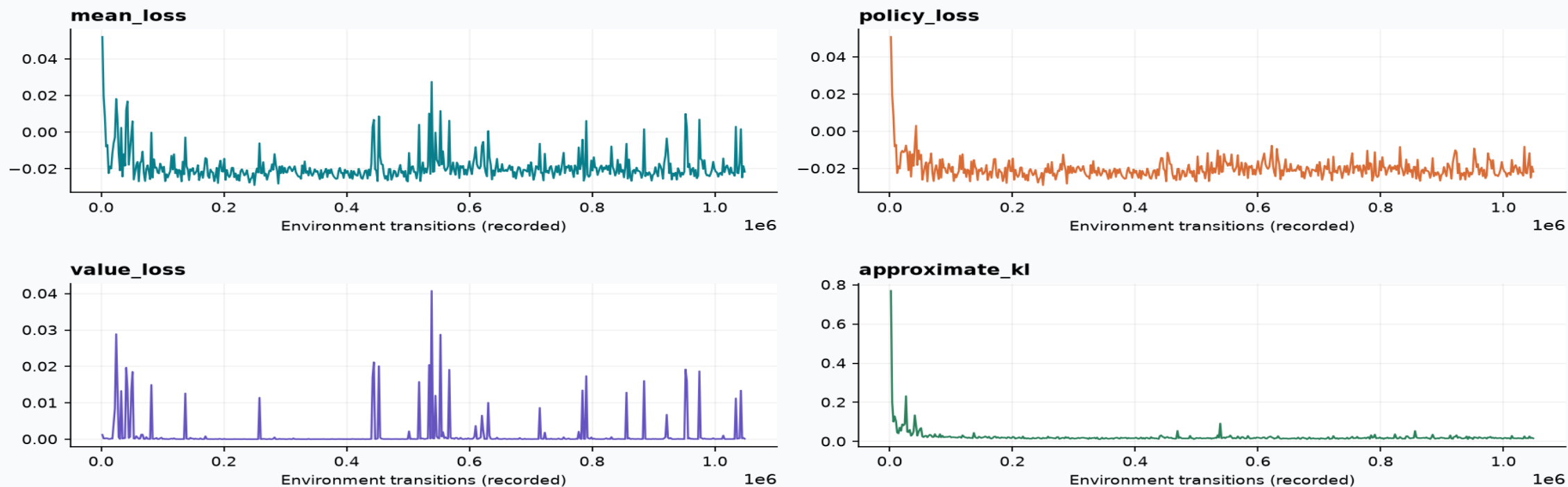
Recorded reward signals — different definitions, not interchangeable



Training reward is an optimization signal, not the acceptance test. See [raw/<case>/](#) for complete recorded fields.

Loss: inspect actor, critic and policy drift

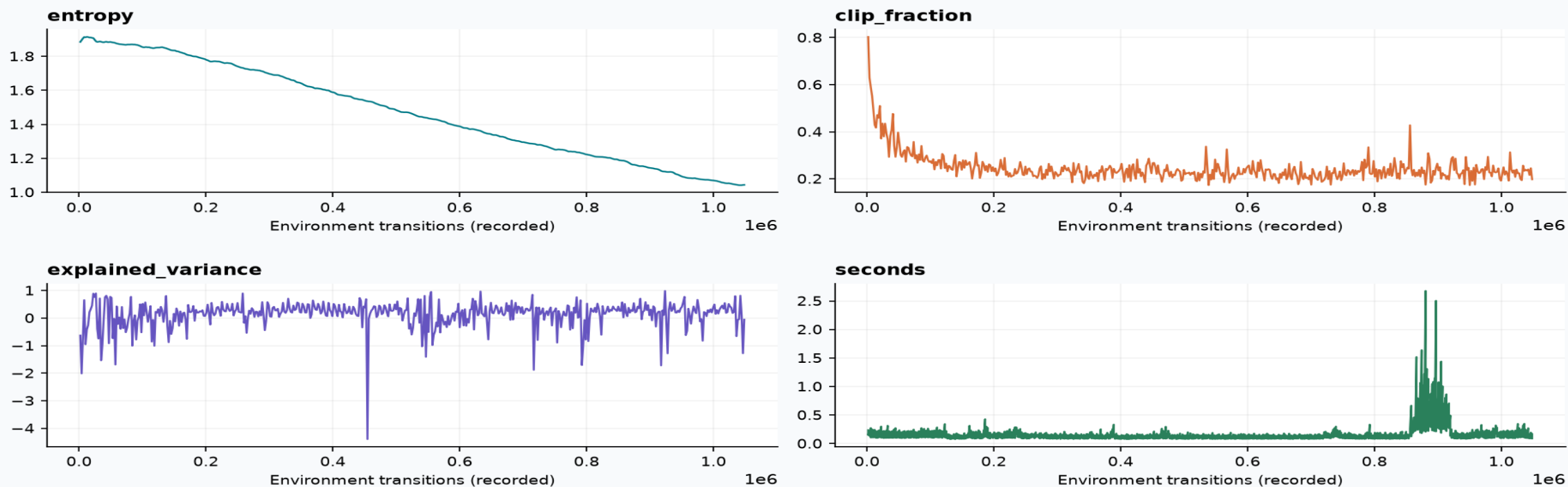
Recorded PPO optimization — raw values, no smoothing



A lower loss does not prove a better robot skill. Loss sign and scale depend on the implementation.

Diagnostics: is the optimizer behaving?

Recorded update diagnostics — absent fields are explicit



Entropy, clipping, explained variance and timing are diagnostics — not a standalone pass/fail certificate.

Metric matrix: numerical audit of this trace · 1/3

- 01 approximate_kl: first 0.7690 → final 0.01645; min 0.01199, max 0.7690; 512 logged samples.
- 02 clip_fraction: first 0.8031 → final 0.1975; min 0.1729, max 0.8031; 512 logged samples.
- 03 entropy: first 1.885 → final 1.044; min 1.042, max 1.915; 512 logged samples.
- 04 explained_variance: first -0.6301 → final -0.05376; min -4.396, max 0.9882; 512 logged samples.

Metric matrix: numerical audit of this trace · 2/3

- 05 mean_loss: first 0.05201 → final -0.02159; min -0.02890, max 0.05201; 512 logged samples.
- 06 mean_raw_return: first 9733 → final 9516; min -1078, max 1.008e+4; 1597 logged samples.
- 07 mean_reward: first 0.03273 → final 0.03256; min 0.03087, max 0.03439; 512 logged samples.
- 08 policy_loss: first 0.05076 → final -0.02173; min -0.02894, max 0.05076; 512 logged samples.

Metric matrix: numerical audit of this trace · 3/3

09

value_loss: first 0.001246 → final 0.0001441; min 0.00002407, max 0.04076; 512 logged samples.

Verification: define success before looking · 1/2

- 01 Require every episode to complete 500 samples without termination, with finite physical metrics and upright fraction ≥ 0.95 .
- 02 Require commanded fraction ≥ 0.50 , directed speed ≥ 0.06 m/s, signed intended displacement ≥ 0.30 m, and speed ratio ≥ 0.30 .
- 03 Require at least 4 alternating support switches, each foot air fraction ≥ 0.05 , contact fraction ≥ 0.10 , and bilateral lift/touch events.
- 04 No both-feet aerial minimum is required; the selected gait's 0.002 aerial fraction is consistent with walking rather than running.

Verification: define success before looking · 2/2

05

Verify the exact experiment identity: 2 cm height, blend 0, 0.014 kg per foot, fixed 0.25 m/s command, and nominal XML settings.

06

Require negative controls to fail; validate hashes, finite JSONL metrics, dynamic batch x 61 to x 14 ONNX, parity, decoded MP4, and contact sheet.

Results: what was actually measured · 1/3

- 01 Actual final evidence: PASS for stilts-e2e-20260907-v3; Studio evaluation passed 16/16 complete 10-second episodes.
- 02 Independent seeds 501-505 passed 5/5; directed speeds were 0.22997129111680586-0.2416003712839324 m/s.
- 03 Signed intended displacement was 1.2806188720775356-2.2887101436867505 m; speed ratios were 0.9198851644672235-0.9664014851357297.
- 04 All five had upright fraction 1.0, 90-92 support switches, left air 0.364-0.410000000000000003, and right air 0.35-0.36.

Results: what was actually measured · 2/3

- 05 Aerial fraction was 0.002 in each five-seed audit episode; no aerial minimum exists for the Stilt gate.
- 06 The v2 initializer already passed comparison seed 501; v3 raised raw return from 10431.73 to 11189.72, a change of 757.9966477797989.
- 07 Zero action fell after 39-46 steps and random action also failed; transient release speed did not satisfy horizon, displacement, or support gates.
- 08 ONNX/native maximum action error was $9.5367431640625e-7$; learned tensor L2 change from the initializer was 6.614700815060063.

Results: what was actually measured · 3/3

09

Final-100 means: policy_loss -0.02061349222087301, value_loss 0.00126573708695787, mean_loss -0.01934775511010912.

10

Final-100 approximate_kl 0.01905187215463229, clip_fraction 0.2322412109375, explained_variance 0.19922025200911597.

Watch the actual saved rollout



BEFORE PLAY

Predict one success signal and one failure signal.

AFTER PLAY

Which claims does this clip support? What remains untested?

10.0 s saved MP4 • Click in PowerPoint Slide Show • Backup: videos/stilts-rollout.mp4

Limits and failure analysis · 1/2

- 01 The evidence covers one fixed 2 cm morphology, blend 0, 0.014 kg mass, and one 0.25 m/s command; morphology generalization is unknown.
- 02 Height, blend, and mass are not observed by the actor, so changing hardware geometry is outside the demonstrated task.
- 03 The base already passed one comparison seed; v3 supports refinement, not sole credit for acquiring stilt locomotion.
- 04 No BAM dynamics, bus latency, randomized mass/friction/noise/delay/yaw, uneven terrain, pushes, or alternative stilt dimensions were tested.

Limits and failure analysis · 2/2

05

Simulation omits attachment compliance, backlash, tolerances, loosening, cable routing, current limits, temperature, battery sag, and impacts.

06

A compatible 61-to-14 ONNX interface is necessary for deployment tooling but is not a physical safety certificate.

Student lab: change one variable, keep the gates · 1/2

- 01 Recommended exercise, not saved evidence: locate observable yaw-rate indices and list morphology variables missing from the actor input.
- 02 Recommended exercise, not saved evidence: compute the weight-20 linear tracking reward for velocity (0.20,0.02,0.01) and command (0.25,0,0).
- 03 Recommended exercise, not saved evidence: explain why moving only the foot site would not reproduce the changed collision geometry.
- 04 Recommended exercise, not saved evidence: verify base and continuation PPO counts and distinguish fresh transitions from epoch reuse.

Student lab: change one variable, keep the gates · 2/2

05

Recommended exercise, not saved evidence: predeclare a held-out matrix of stilt heights, blends, and masses with pass/fail interpretation.

Submit an experiment, not just a video

- 01 Before training: write one hypothesis, the changed parameter, the fixed baseline, seed list, acceptance criteria and time/transition budget.
- 02 During training: save the exact command, environment versions, initialization, checkpoint and raw reward/loss/KL metrics. Never overwrite the saved reference run.
- 03 After training: evaluate the frozen exported policy on matched seeds; include all failures, aggregate metrics, the full rollout video and a source-bound result table.
- 04 Rubric / 10: reproducibility 2; environment and reward explanation 2; PPO and metrics interpretation 2; honest verification 2; one evidence-based next experiment 2.

Source map and further study · 1/2

01

AGENTS.md

02

microduck_local/AGENTS.md

03

microduck_local/src/microduck_local/contract.py

04

README.md

Source map and further study · 2/2

05

`docs/microduck-training-student-book/chapters/02a-ppo-code-lab.md`

06

`docs/microduck-training-student-book/chapters/02b-gae-workshop.md`

07

`scripts/scenarios/README.md`

08

`README.md`